Consensus Recommendations on the Aesthetic Usage of Botulinum Toxin Type A in Asians

Bong Kyun Ahn, MD,* Youn Sung Kim, MD,† Hong Jig Kim, MD,‡ Nark Kyoung Rho, MD,§ and Hei Sung Kim, MD¶

BACKGROUND The use of botulinum toxin has rapidly expanded into various aesthetic applications. Any guideline representing a consensus for aesthetic treatments using botulinum toxin type A (BTA) in Asians has not been published.

OBJECTIVES To provide consensus recommendations on common aesthetic problems which are treated by neurotoxin in Asians.

METHODS A panel of experienced Korean dermatologists was convened to develop a clinical consensus on common aesthetic problems involving the face, neck, and calves in Asians, based on their own extensive experience.

RESULTS The consensus recommendations address general questions regarding treatment and provide specific guidelines on each common aesthetic indication. The recommended final concentration of BTA was 50 U/mL after reconstitution with physiologic saline. For horizontal forehead lines, the members recommended nine injections in two rows into the frontalis with 1 U/point. For glabellar lines, the members recommended three injection points (a total of 8 U). For crow's feet, the members recommended three injections per side (7 U/side) at the lateral part of the orbicularis oculi. For infraorbital wrinkles, one to two points per side in the superficial subcutaneous space approximately 1 cm below the lash line were recommended (1–2 U/side). For nasal flare, one injection point in the middle of each ala nasi was recommended (a total of 2 U). For depressed nasal tip, a single injection deep within the columella was recommended, with a dose of 3 U. For benign masseter hypertrophy, the members recommended a six-point injection to the masseter (three points per side for a total of 50–60 U). For the treatment of calf hypertrophy, the members recommended a total dose of 100 to 120 U (50–60 U/side), divided between six injection points (approximately 8–10 U/point).

CONCLUSION This guideline provides a framework for physicians who wish to perform safe and efficacious injections of BTA in Asians.

The authors have indicated no significant interest with commercial supporters.

Injection with botulinum toxin type A (BTA) is one of the most widely performed noninvasive cosmetic procedures in the world. The first medical use of BTA was reported in 1980,[1] and the first cosmetic use of BTA—as a safe and effective treatment for glabellar lines—was described in a seminal paper published in 1992.[2] The Food and Drug Administration granted approval for the treatment of facial wrinkles on April 12, 2002. Consensus recommendations for onabotulinumtoxinA (Botox, Allergan, Inc., Irvine, CA) from the United States[3] and for abobotulinumtoxinA (Dysport, Ipsen Ltd, Slough, UK) from Europe[4,5]

*MYD Aesthetic Laser and Cosmetic Surgery Center, Seoul, Korea; †Kimbelle Catholic Skin Clinic, Daejeon, Korea; ‡OhKim's Skin and Laser Clinic, Seoul, Korea; §Leaders Aesthetic Laser and Cosmetic Surgery Center, Seoul, Korea; ¶Incheon St. Mary's Hospital, Incheon, Korea

Special thanks to Chonghyeok Kim, Deug Gon Moon, Ee Seok Lim, Geun Soo Lee, Hangrae Cho, Hyeon Sook Lee, Jae Young Hwang, Kwang Ho Han, Sang Duck Kim, Sang Ju Lee, Suk Bae Seo, and Woo Sun Lee.

© 2013 by the American Society for Dermatologic Surgery, Inc. • Published by Wiley Periodicals, Inc. •
ISSN: 1076-0512 • Dermatol Surg 2013;39:1843–1860 • DOI: 10.1111/dsu.12317

began to arise in 2004, and revised[6] or updated[7–9] consensus recommendations have recently been published to help physicians treat Caucasians. In Asian countries, the use of BTA in cosmetic procedures started when the commercial preparation of BTA became available, roughly 1999.[10] With ethnic differences in skin texture, facial muscle anatomy, pharmacologic reactions, and notions of beauty, it is difficult to apply consensus recommendations based on Caucasians directly to Asians. Because there are no published regional guidelines on BTA injection for cosmetic purposes in Asians, we believe that our general guidelines will be beneficial in treating Asians.

Considerations Regarding BTA Formulations

Commercial preparations of BTA products that are available in Korea are Botox, Botulax (Hugel Inc., Chuncheon, Korea), BTX-A (Hugh Source, Hong Kong, China), Dysport, Neuronox (Medytox Inc., Seoul, Korea), and Xeomin (Merz Pharma GmbH & Co. KGaA, Frankfurt/Main, Germany) (in alphabetical order). Of these, only Dysport is abobotulinumtoxinA. It has different bacterial strains of origin, different methods of purification and stabilization, different chemical and biological properties, and different units (Speywood units) of potency. This product has mainly been used for facial aesthetics in Europe, whereas onabotulinumtoxinA products have most commonly been used in the United States.[7] The dose equivalency of the products (Botox: product) are Botulax, 1:1; BTX-A, 1:1; Dysport, 1:2–1:4; Neuronox, 1:1; and Xeomin, 1:1.[11–13] Botox and Neuronox are the most frequently used products in Korea, and as mentioned, their potencies are thought to be almost identical.[13] The Korean Academy of Corrective Dermatology (KACD) consensus recommendation is described with Botox and Neuronox as standard BTA products.

Methods of Consensus Development

KACD consists of 17 dermatologists who have extensive experience in the cosmetic usages of botulinum toxin in Asians. Members have administered botulinum toxin for an average of 9.4 years (range 7–13 years) to a variety of patients: Koreans (55%), Chinese (30%), Japanese (10%), and others (5%). Board members convened to develop consensus recommendations on various cosmetic indications of the face, neck, and calves based on their own experience (Table 1). A strong consensus was defined as approval from at least 90% of the board members.

Consensus Recommendations

Reconstitution

In a questionnaire-based study conducted by a Korean BTA company (Medy-Tox, Inc.) in 2009, 500 Korean physicians (300 dermatologists, 200 plastic surgeons) each diluted a vial of BTA (100 U) with 2.5 mL of saline (49%) followed by 2.0 mL (21%) and 3.0 mL (14%) for horizontal forehead and glabellar wrinkles and benign masseter hypertrophy. We recommend reconstituting a vial of BTA (100 U) in preservative-free 0.9% saline (2.0 mL) Recommended indications are arranged in cephalocaudal order.

Horizontal Forehead Lines

Horizontal forehead lines are one of the most common indications for BTA injection in Asians.

Injection Target, Points, Dose, and Technique The primary muscle of the horizontal forehead lines is the frontalis muscle (Figure 1). Draw an imaginary horizontal line 2 cm above the orbital rim and inject at five regularly spaced points. Then, 1 to 1.5 cm above the first line, draw an imaginary line and inject at four points in the same manner (Figure 2). Inject 1 to 1.5 U at each point. (Higher doses may be needed for men.) The total recommended dose range is 6 to 13.5 U. Intradermal and oblique injection is recommended for Asians.

Ethnic Considerations Several articles report ethnic differences in the contractility and shape of the frontalis muscle. With regard to forehead contractility, Tzou and colleagues[14] showed that

TABLE 1. Korean Academy of Corrective Dermatology Consensus Recommendations on the Aesthetic Usage of Botulinum Toxin Type A in Asians

	Target Muscle	*Injection Level*	*Injection Points,* n	*Injection Dose, U*	*Total Dose, U*
Horizontal forehead lines	Frontalis	Intradermal	6–9	1–1.5	6–13.5
Glabellar lines (moderate)	Corrugator Procerus	Intramuscular	3	4 2	8
Glabellar lines (severe)	Corrugator Procerus	Intramuscular	5	4 2	12
Brow lifting	Brow depressors Procerus Corrugator Depressor supercilii Orbicularis oculi	Intramuscular	4–5	1 2 2–4 (optional)	6–10
Crow's feet	Orbicularis oculi	Intradermal, subcutaneous, or intramuscular	6 (3/side)	3 2	14 (7/side)
Infraorbital wrinkles	Orbicularis oculi	Intradermal	2–4	0.5	1–2
Bunny lines	Nasalis (transverse part)	Intradermal, subcutaneous	3	2	6
Nasal flare	Dilator naris	Intramuscular	2	1	2 (1/side)
Nose tip lifting	Depressor septi nasi	Intramuscular	1–3	3 2 (optional)	7–10
Perioral wrinkles	Orbicularis oris	Intradermal (strictly)	4	0.2–0.3	0.8–1.2
Gummy smile	Upper lip elevators Levator labii superioris Alaeque nasi Levator labii superioris Zygomaticus minor	Intramuscular	2	2–3	4–6
Marionette lines	Depressor anguli oris	Intramuscular	2	3–5	6–10
Cobblestone chin (mild)	Mentalis	Intramuscular	1	5	5
Cobblestone chin (severe)	Mentalis	Intramuscular	2 points	5	10
Masseter hypertrophy	Masseter	Intramuscular	6 (3/side)	8–10	48–60 (24–30/side)
Platysmal bands	Platysma	Intramuscular	3–4/band	2	24–32
Calf contouring	Gastrocnemius	Intramuscular	12 (6/side)	8–10	100–120 (50–60/side)

Europeans have larger facial movements in the frontal region than Asians. Tsukahara and colleagues[15] found significantly higher wrinkle scores in Caucasians than in Japanese. In terms of shape, Koreans were shown to have wider foreheads than white North Americans.[16] Ethnic differences in the frontalis may influence the method of BTA treatment,[17] which is summarized in Table 2.

Safety Concerns All injections should be 1 to 2 cm above the orbital rim to minimize the risk of brow ptosis. In cases in which the external portion of the frontalis muscle is not treated sufficiently, excessive elevation of the external third of the eyebrow can be seen (the "Samurai face").

Glabellar LINES

The glabella is the most common site for BTA injection in Asians.

Injection Target, Points, Dose, and Technique The primary muscles of the glabellar complex are the

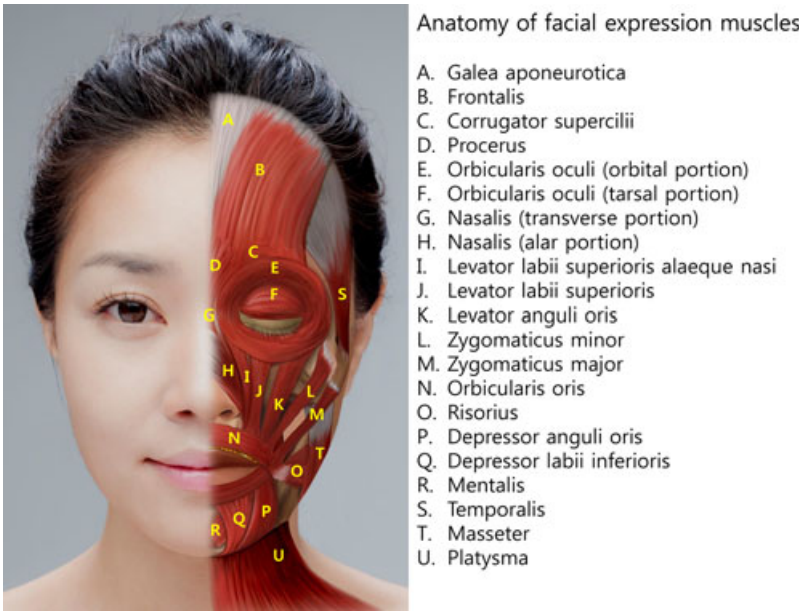
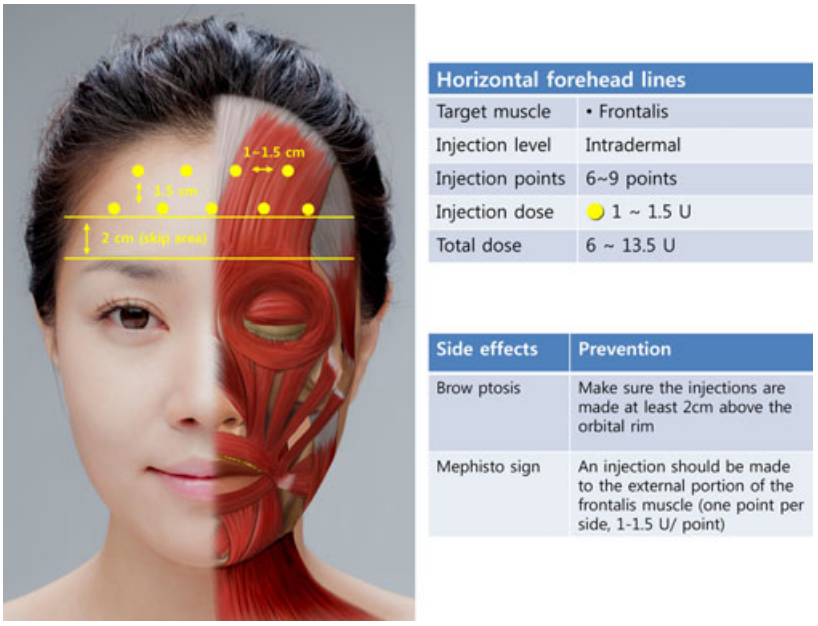


Figure 1. Anatomy of facial expression muscles.



Horizontal forehead lines	
Target muscle	• Frontalis
Injection level	Intradermal
Injection points	6~9 points
Injection dose	1 ~ 1.5 U
Total dose	6 ~ 13.5 U

Side effects	Prevention
Brow ptosis	Make sure the injections are made at least 2cm above the orbital rim
Mephisto sign	An injection should be made to the external portion of the frontalis muscle (one point per side, 1-1.5 U/ point)

Figure 2. Botulinum toxin injection to the horizontal forehead lines.

corrugator supercilii, the procerus, and the depressor supercilii (Figure 1). A three-point injection is recommended for glabellar lines. Draw an imaginary line between the origin of the medial brow and the contralateral inner canthus and inject 4 U of BTA at the point where these lines intersect (Figure 3). Two units of BTA are injected at the intersection of the vertical line above the inner canthus and the brow line. In patients with large corrugator muscles, five injection points can be used (Figure 4). The procerus is a superficial muscle and should not be injected too deeply. In contrast, the corrugator supercilii muscles

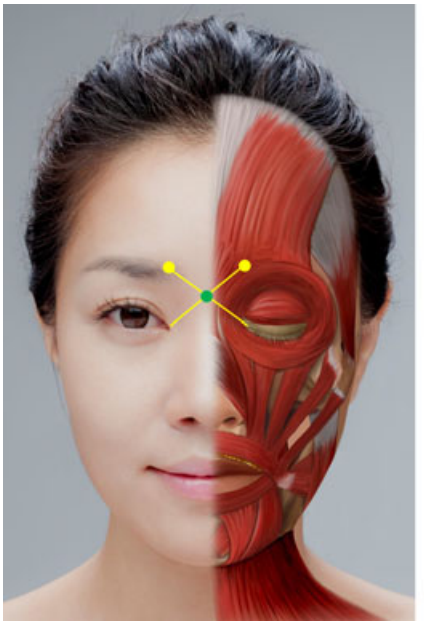
TABLE 2. Indications for Which the Korean Academy of Corrective Dermatology (KACD) Consensus Recommendations Differ Significantly from Western Recommendations

	KACD	*Carruther (2004)3*	*Carruther (2008)6*	*Kane (2010)7*	*Ascher (2010)8*
KACD indications that show the greatest differences in total BTA dose compared to other recommendations					
Glabellar lines	8 U	20–30 U in women 30–40 U in men	10–30 U in women 20–40 U in men	30–70s.U* in women 50–80s.U* in men	50s.U*
Crow's feet	7 U/side	6–15 U/side	5–15 U/side in women 10–15 U/side in men	20–60s. U*/side	13–30s.U*/side
Perioral wrinkles	0.8–1.2 U	5–6 U	4–5 U	2.5–16 s.U*	4–12 s.U*
KACD indications in which the recommended level of injection is different from other recommendations					
Horizontal forehead lines	Intradermal	Intramuscular	Intramuscular	Not specified	Intramuscular
Crow's feet	Intradermal	Intramuscular	Intramuscular	Intradermal	Intradermal
KACD indications for which the number of injection points differ from other recommendations					
Horizontal forehead lines	9 points	4–8 points	4–8 points	4 points	4–6 points
Glabellar lines	3 points (5 points in severe case)	5–7 points	5–7 points	5 points	5 points
KACD indications in which the site of injection differs from other recommendations					
Infraorbital wrinkles	1 cm below the ciliary line	Not specified	1–2 mm below the ciliary line	Not specified	2 mm below the ciliary line
Brow lift	Corrugator and eyebrow tail	Not specified	Not specified	Not specified	Forehead and eyebrow tail
KACD indications that have not been commonly mentioned in prior recommendations					
Brow lift	Recommended	Not specified	Not specified	Not specified	Recommended (only lateral eyebrow)
Nasal flare	Recommended	Not specified	Not specified	Not specified	Not specified
Nose tip lifting	Recommended	Not specified	Not specified	Not specified	Recommended
Gummy smile	Recommended	Not specified	Just introduced as a new technique	Recommended	Not specified
Masseter hypertrophy	Recommended	Not specified	Just introduced as a new technique	Not specified	Recommended
Calf contouring	Recommended	Not specified	Not specified	Not specified	Not specified
Indications not recommended by KACD, unlike other recommendations					
Perioral wrinkles (lower lip treatment)	Not recommended	Recommended	Recommended	Recommended	Recommended

*Direct comparison not possible because Speywood units (s.U) and U are different units for quantifying potency.

lie deep relative to the frontalis muscle and should be injected deeply (closer to the periosteum).

Ethnic Considerations The results of a study with Chinese cadavers suggested that a large number of Asians have "short, square type" corrugator muscles.[18] In contrast, Caucasians often have long, transversely developed corrugators.[19] These ethnic differences may explain the more-conservative approach needed in Asians when treating



Glabellar frown lines (moderate)	
Target muscle	• Corrugator • Procerus
Injection level	Intramuscular
Injection points	3 points
Injection dose	2 U 4 U
Total dose	8 U

Side effects	Prevention
Eyelid ptosis	Avoid injecting high dosage and inject slowly. Gently press the eyeball with a free finger to avoid diffusion of BTA into the orbital area.
Medial brow elevation	Make an injection to the frontalis muscle 3cm above the affected eyebrow (one point per side, 2 U/ point)

Figure 3. Botulinum toxin injection to the glabellar frown lines (moderate).

glabellar wrinkles with BTA (Table 2) (8-U injection in Asians vs 20- to 30-U injection in Caucasians).[20]

Safety Concerns Upper eyelid ptosis can be the most serious complication. A three-point injection can cause medial brow elevation.

Brow Lifting

Brow lifting is not a popular procedure in Asia but is at times performed in conjunction with other upper facial BTA treatments.

Injection Target, Points, Dose, and Technique The target muscles for brow lifting are the brow depressors, which include the procerus, the corrugators, the depressor supercilii, and parts of the orbicularis oculi (Figure 1). A four-point injection (2 points per side) is recommended (Figure 5). A single injection should be made into the pars orbicularis at the tail of each eyebrow. The recommended dose is 1 U per injection point. Two additional injections should be made into the corrugators; 2 U per injection point is recommended. The total dose range for brow lifting is 6 to 10 U. If injection into the median portion does not provide sufficient brow lifting, an optional 2- to 4-U injection into the procerus can be performed. We recommend an intramuscular injection for the eyebrow tail (superficial) and corrugators (deep and perpendicular).

Ethnic Considerations Previous recommendations have rarely mentioned "brow lifting," with the exception of one article from Europe, in which the authors emphasized lifting only the lateral portion of the brow (Table 2).[8] This difference stems from the disparity in perception of a desirable and beautiful brow shape between Asians and Caucasians. In an article involving digital modification of photographs, the ideal brow peak was located above the lateral canthus in a white model, whereas it was more medial in a young Asian model.[21] Asian women do not favor an abruptly arched brow with a lateral peak because it is felt to appear aggressive. The KACD recommendation includes suppression of corrugators to achieve a favorable Asian brow shape.

Safety Concerns This treatment is usually effective and safe. Eyelid and brow ptosis occur rarely.

Crow's Feet

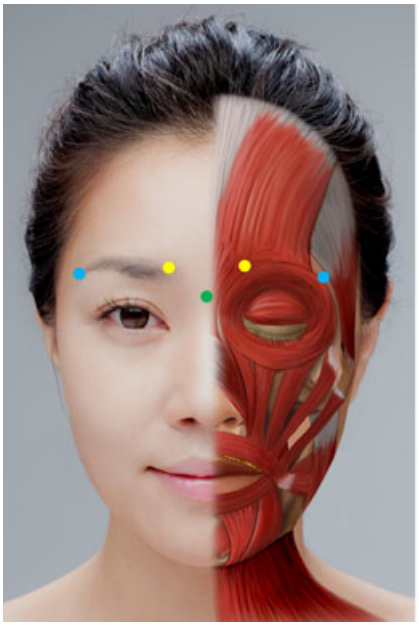
BTA injection for crow's feet is one of the most popular cosmetic procedures in Asia.



Glabellar frown lines (severe)	
Target muscle	• Corrugator • Procerus
Injection level	Intramuscular
Injection points	5 points
Injection dose	4 U 2 U
Total dose	12 U

Side effects	Prevention
Eyebrow ptosis	Take care when injecting on points 1cm above the eyebrow to avoid BTA diffusion to the frontalis. Avoid high dosages and do not inject superficially.
Eyelid ptosis	Avoid injecting high dosage and inject slowly. Gently press the eyeball with a free finger to avoid diffusion of BTA into the orbital area.
Medial brow elevation	Make an injection to the fontalis muscle 3cm above the affected eyebrow (one point per side, 2 U/ point)

Figure 4. Botulinum toxin injection to the glabellar frown lines (severe).

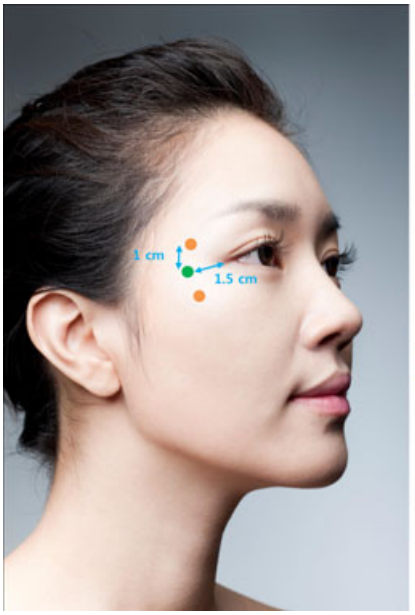


Brow lifting	
Target muscle	• Brow depressors - Procerus - Corrugator - Depressor supercilii - Orbicularis oculi
Injection level	Intramuscular
Injection points	4-5 points
Injection dose	1 U 2 U 2 ~ 4 U (optional)
Total dose	6-10 U

Side effects	Prevention
Brow ptosis	In a hyperkinetic patient, avoid high dosage injection to the lateral part of the orbicularis oculi.

Figure 5. Botulinum toxin injection for brow lifting.

Injection Target, Points, Dose, and Technique The primary muscle in crow's feet is the external part of the orbicularis oculi (Figure 1). For the treatment of crow's feet, injection to the canthal region (three points per side) is recommended (Figure 6). The first injection is usually made approximately 1.5 cm lateral to the outer canthus (dose 3 U), with the other two injection points (2 U each)



Lateral canthal wrinkles (Crow's feet)	
Target muscle	• Orbicularis oculi
Injection level	Intradermal, or Subcutaneous, or Intramuscular
Injection points	6 points (3points/side)
Injection dose	3 U 2 U
Total dose	14 U (7 U/side)

Side effects	Prevention
Upper lip ptosis (rare)	BTA injection to the mid-face region may affect the zygomaticus major, a muscle involved in lifting the mouth corners. Avoid injecting below the orbicularis oculi sector
Bruises	Avoid injecting too deep. Compress after injection.

Figure 6. Botulinum toxin injection to the lateral canthal wrinkles (crow's feet).

approximately 1 cm above and below the primary injection point. An intradermal injection is preferred, which minimizes the risk of bruising while retaining effectiveness.

Ethnic Considerations In general, the degree of periorbital wrinkling is higher in Caucasians than Asians when one smiles fully.[14,22] By comparing two studies from Europe and Japan that present fundamental data on facial muscle fibers, we could surmise that the muscle fiber size of orbicularis oculi is smaller in Asians than in Europeans. Skin thickness is another important factor determining the degree of periorbital wrinkling, with Asians shown to have thicker facial skin (and a thicker epidermis) than Caucasians.[23] Although insufficient, these data help explain the ethnic difference in recommendation (Table 2) (7 U/side, intradermal injection in Asians vs 10–15 U/side, intramuscular injection in Caucasians).

Safety Concerns The most frequent adverse event is mild periorbital hematoma. Deep injection or injection into the inferior region of the zygomaticus major should be avoided becasue it may cause unwanted effects such as drooping mouth corners.

Infraorbital Wrinkles

Intraorbital wrinkles are commonly treated using BTA in Asia.

Injection Target, Points, Dose, and Technique The primary muscle of the infraorbital wrinkle is inferior horizontal fibers of the orbicularis oculi (Figure 1). BTA (0.5 U) should be injected into the superficial subcutaneous space approximately 1 cm below the lash line (Figure 7). One to two points per side is recommended, with a total dose of 1 to 2 U. Oblique and intradermal injection is recommended.

Ethnic Considerations There are significant racial differences in perceptions of a groove that is caused by contraction and hypertrophy of the pretarsal orbicularis oculi (the "jelly roll"). For Caucasians, the 'jelly roll' is thought to make them look dull, and many opt for rounder eyes, particularly when smiling. Thus, many have the desire to remove the "jelly roll." Carruthers and colleagues[6] introduced a



Infraorbital wrinkles	
Target muscle	• Orbicularis oculi
Injection level	Intradermal
Injection points	2 ~ 4 points
Injection dose	0.5 U
Total dose	1 ~ 2 U

Side effects	Prevention
Fat pad bulging	Patients who are too old, or who show significant lower orbital fat sagging and severe infraorbital skin laxity should be avoided.
Disappearance of the "charming roll"	Avoid injecting 1cm within the lower eyelid margin and injecting large doses
Bruises	Use a 31 G needle. Inject superficially.

Figure 7. Botulinum toxin injection to the infraorbital wrinkles.

method to treat the "jelly roll" using BTA injection. For Asians, this structure is thought to make women look younger and cute and is thus referred to as the "charming roll." Many Asian patients wish to increase the size of this roll through filler injection. These differences in the perception of beauty explain the ethnic difference in recommendations (Table 2) (injection of BTA 1 cm below the ciliary line in Asians vs 1–2 mm below the ciliary line in Caucasians).

Safety Concerns The lower orbital fat pad may bulge because of the reduction of infraorbital muscle tone after BTA injection. Patients should also be warned of the possibility of infraorbital puffiness before treatment.

Bunny Lines

BTA injection for bunny lines is not uncommon in Asia.

Injection Target, Points, Dose, and Technique Bunny lines result from contraction of the transverse portion of the nasalis (Figure 1). A three-point injection is recommended for treatment (1 point in the middle and 1 on each side) (Figure 8). The recommended total dose is 6 U (2 U per injection point). The injections should be performed superficially.

Ethnic Considerations The total dose in the KACD recommendation is slightly higher than those of Western recommendations.

Safety Concerns Upper lip ptosis can occur when the levator labii superioris alaeque nasi is blocked.

Nasal Flare and Depressed Nasal Tip

Nasal flare and depressed nasal tip are frequent indications for BTA in Asian patients.

Injection Target, Points, Dose, and Technique The contraction of the inferior portion of the nasal muscle called the dilator naris causes nasal flare.[24] The nasal septum depressor muscle is involved in ptosis of the nasal tip.[25] One injection at the middle of each ala nasi is sufficient for nasal flare (Figure 9). The recommended dose is 1 U per injection point. Perpendicular intramuscular injection is recommended. One deep intramuscular injection at the



Nasal wrinkles (Bunny lines)	
Target muscle	• Nasalis (transvers part)
Injection level	Intradermal Subcutaneous
Injection points	3 points
Injection dose	2 U
Total dose	6 U

Side effects	Prevention
Upper lip ptosis	Avoid injecting downwards and outwards near the nasal ala- This may cause paralysis of the levator labii superioris alaeque nasi which lifts the mouth corners.

Figure 8. Botulinum toxin injection to the nasal wrinkles (bunny lines).



Nasal flare	
Target muscle	• Dilator naris
Injection level	Intramuscular
Injection points	2 points
Injection dose	1 U
Total dose	2 U (1 U/side)

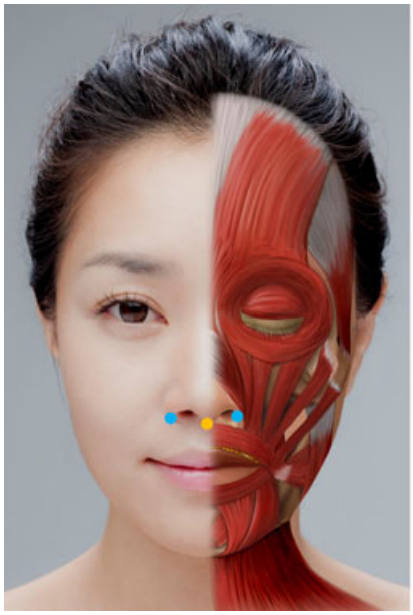
Side effects	Prevention
Upper lip ptosis	Avoid injecting outwards. Inject small doses, intramuscularly to the ala nasi

Figure 9. Botulinum toxin injection for correction of nasal flare.

base of the columella is recommended for elevation of the nasal tip (Figure 10). The recommended dose is 3 U. If necessary, additional injections (2 U) can be made into each side of the nasal muscle nasi.

Ethnic Considerations Because many Asians prefer a "Caucasian nose," rhinoplasty is one of the most popular and frequently performed procedures in Asia. Asians typically have lower nasal bridges, bulbous and poorly projected tips, a small septum, and wide nostrils. Because nasal flare and depressed nasal tip make the nose look flatter and wider, the demand for such correction is higher in Asians than Caucasians.

Safety Concerns Injections to the ala nasi must be performed with care because there is a risk of lowering



Nose tip lifting	
Target muscle	• Depressor septi nasi
Injection level	Intramuscular
Injection points	1~3 points
Injection dose	3 U 2 U (optional)
Total dose	7 ~ 10 U

Side effects	Prevention
Upper lip ptosis	For patients who are expected to have difficulty in making full smiles, give a single injection on the central portion (depressor septi nasi)
Slurring and drooling	To avoid BTA diffusion into the orbicularis oris, an injection should be made close up to the nasal tip

Figure 10. Botulinum toxin injection for nose tip lifting.

of the corners of the mouth, causing lip ptosis and lengthening the philtrum. Impairments of labial function, such as speech impairments and difficulties in eating and drinking with a straw, can occur.

Perioral Wrinkles

BTA or filler treatment for perioral wrinkles is less frequently performed in Asia than in Western countries.

Injection Target, Points, Dose, and Technique The target muscle for perioral wrinkles is the orbicularis oris (Figure 1). Four symmetrical injections on the upper lip are recommended for perioral wrinkles (Figure 11). The injection points should be at the vermilion border, with the lateral points at least 1 cm away from the mouth corners. The recommended total dose for the perioral area is approximately 0.8 to 1.2 U. The injections should be intradermal and as superficial as possible.

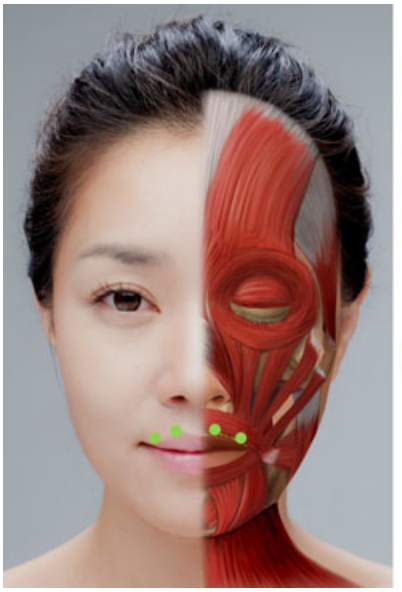
Ethnic Considerations Morphologically, a high percentage of Asians have protruding lips, which is rare in Caucasians. This protruding oral structure is thought to make Asians more sensitive to complications of BTA treatment. The ethnic difference in recommendation is shown in Table 2 (1-U injection in Asians vs 5 U in Caucasians).

Safety Concerns Difficulty in pursing the lips can occur with excessive doses or deep injection of BTA in the perioral area. Speech impairments and difficulties eating and drinking with a straw can also occur.

Gummy Smile

Many Asians consider People with a "gummy smile" to be unsightly. The procedure requires skill, and only highly experienced practitioners should perform it.

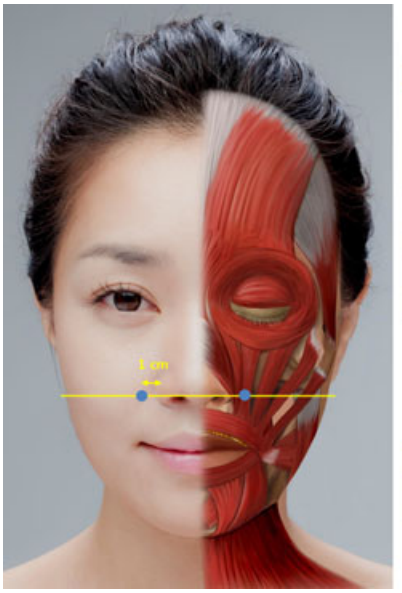
Injection Target, Points, Dose, and Technique A "gummy smile" can be corrected by injecting BTA to the upper lip elevators, such as the levator labii superioris alaeque nasi, levator labii superioris, and zygomaticus minor (Figure 1). Injection of 2 to 3 U of BTA is performed at the nasofacial groove, which is usually approximately 1 cm lateral to the nasal ala (Figure 12). Injection should be perpendicular, intramuscular and deep (just above the periosteum).



Perioral wrinkles (Smoker's lines)	
Target muscle	• Orbicularis oris
Injection level	Intradermal (strictly)
Injection points	4 points
Injection dose	0.2 ~ 0.3 U
Total dose	0.8 ~ 1.2 U

Side effects	Prevention
Slurring or drooling	Inject small doses, superficially. Injections should be made close up to the vermilion border. It is advised to explain the possibility of slight drooling to the patient prior treatment.

Figure 11. Botulinum toxin injection to perioral wrinkles (smoker's lines).



Gummy smile (Gingival show)	
Target muscle	• Upper lip elevators - Levator labii superioris Alaeque nasi - Levator labii superioris - Zygomaticus minor
Injection level	Intramuscular
Injection points	2 points
Injection dose	2 ~ 3 U
Total dose	4 ~ 6 U

Side effects	Prevention
Upper lip ptosis	Avoid injecting high doses. Inject on the exact points.

Figure 12. Botulinum toxin injection for correction of gummy smile (gingival show).

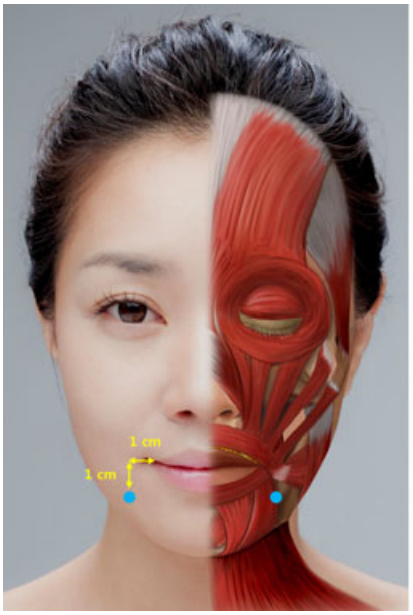
Ethnic Considerations The dose of BTA and injection points recommended by KACD are similar to those in Western recommendations.

Safety Concerns Potential risks associated with treatment of "gummy smile" are upper lip ptosis, excessive lengthening of the philtrum, and drooping of the mouth corners.

Marionette Lines

The use of BTA alone in treating marionette lines is regarded as difficult and risky. It is therefore not commonly performed in Asia.

Injection Target, Points, Dose, and Technique Extensive contraction of the depressor anguli oris and some fibers of the platysma can pull



Marionette lines	
Target muscle	• Depressor anguli oris
Injection level	Intramuscular
Injection points	2 points
Injection dose	3 ~ 5 U
Total dose	6 ~ 10 U

Side effects	Prevention
Slurring and drooling	Ask the patient to droop their mouth corners to the miximum- the practitioner should use his or her fingertip to detect muscle contraction and locate the precise injection point. This prevents BTA diffusion into the Orbicularis oris or depressor labii inferioris

Figure 13. Botulinum toxin injection to marionette lines.

the corners of the mouth downward (Figure 1). A two-point injection into the depressor anguli oris (1 point per side) is recommended, with a total dose of 6 to 10 U (3–5 U per point) (Figure 13). The injection should be intramuscular and perpendicular.

Ethnic Considerations The dose of BTA and injection points recommended by KACD are similar to those in Western recommendations.

Safety Concerns Drooling and speech impairment can occur when the levator anguli oris is affected.

Dimpled Chin

BTA is the most effective treatment for this problem and is commonly performed in Asians. For best results, BTA is often combined with filler augmentation.

Injection Target, Points, Dose, and Technique Dimpled chin is present in people who have a form of mentalis muscle with significant numbers of fibers inserting into the dermis. A midline injection (5 U) is often sufficient to soften a dimpled chin (Figure 14), but for severe cases, a two-point injection close to the center (5 U per point) may be considered (Figure 15). The injection points should be closer to the mandible than the lower lip to avoid diffusion of BTA to the orbicularis oris. Because the mentalis is a deep muscle, the injection should be perpendicular to the skin and should be deep and intramuscular.

Ethnic Considerations The dose and injection points of the KACD recommendation are similar to those in Western recommendations.

Safety Concerns Drooling, speech impairment, mouth asymmetry, and lower lip ptosis can occur.

Benign Masseter Hypertrophy

This indication has a high demand in Asia, and good results can be achieved with repeated injections of BTA.

Injection Target, Points, Dose, and Technique The target muscle is the masseter (Figure 1). A six-point injection into the masseter (3 points per side) is recommended (Figure 16). Physicians should ask patients to clench their teeth to set the boundary of the safe zone. The safe zone is a rectangular area defined by linking the lower border of the mandible,



Cobblestone chin (Mild)	
Target muscle	• Mentalis
Injection level	Intramuscular
Injection points	1 points
Injection dose	5 U
Total dose	5 U

Side effects	Prevention
Slurring and drooling	Make sure to inject BTA in the lower half of the chin- this prevents diffusion of the toxin to the orbicularis oris

Figure 14. Botulinum toxin injection to cobblestone chin (mild).



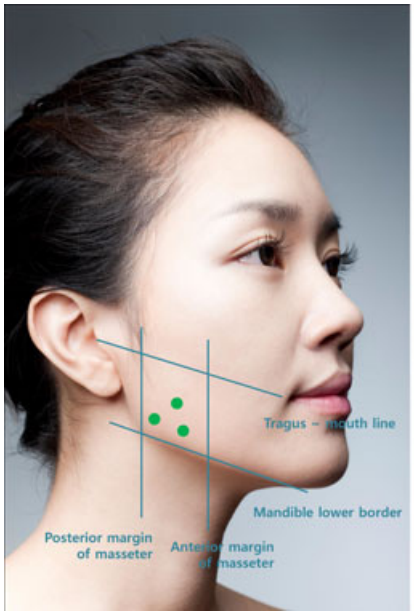
Cobblestone chin (Severe)	
Target muscle	• Mentalis
Injection level	Intramuscular
Injection points	2 points
Injection dose	5 U
Total dose	10 U

Side effects	Prevention
Slurring and drooling	Make sure to inject BTA in the lower half of the chin- this prevents diffusion of the toxin to the orbicularis oris

Figure 15. Botulinum toxin injection to cobblestone chin (severe).

the tragus–mouth corner line, and the anterior and posterior borders of the masseter. Injection into this zone is highly effective and safe. The first injection is administered into the thickest (middle) portion of the masseter muscle 1.5 cm from the border of the mandibular angle. Two other points are located anterior and posterior to the center. Together, the three points form a triangle. A total dose of 50 to 60 U is recommended (8–10 U per point).

Ethnic Considerations Asians, including Koreans, tend to have more-developed mandibles than Westerners. Data show that the distance between the gonions is 12 to 20 mm longer in Korean woman



Benign masseter hypertrophy	
Target muscle	• Masseter
Injection level	Intramuscular
Injection points	6 points (3points/side)
Injection dose	8 ~ 10 U
Total dose	48 ~ 60 U (24 ~ 30 U/side)

Side effects	Prevention
Dimpling	This occurs when too much toxin is injected too deeply. Try to inject superficially.
Sunken cheek	Avoid injecting people with little facial fat. Inject small doses
Pain on mouth opening	Injecting BTA near the mandibular angle may cause toxin diffusion to the pterygoid. Keep the injection points within the safety zone

Figure 16. Botulinum toxin injection to benign masseter hypertrophy.

than in Caucasians.[26] In addition, benign hypertrophy of the masseter muscle is common in Asians and contributes to an undesirable quadrangular and wide lower face. There is high demand for a slim mandible in Asians, and the use of BTA to improve a square jaw appearance is common.

Safety Concerns The risk of significant adverse events is virtually nil, but crunching power has been reported to be reduced for 2 to 4 weeks after injection.[27] There have been a number of complaints of sunken cheeks in slim patients with prominent zygomas. Disappearance of facial dimples and protrusion of muscles with crunching motion occur occasionally.

Platysmal Bands

BTA injection is less frequently performed in Asians than Caucasians for the platysmal bands.

Injection Target, Points, Dose, and Technique The platysma is a thin, broad muscle that originates from the border of the lower jaw and extends to the clavicular region (Figure 1). The location of the platysma muscle can be verified by asking the patient to grimace and tighten the muscle. Once the platysmal bands become apparent, the physician can pinch the bands of muscle with the nondominant hand and inject BTA directly into the muscle band at points approximately 1 to 1.5 cm apart (2 U per site) (Figure 17). Three to four injections are usually made per band, with a total dose of 24 to 32 U. Injections should be performed superficially and intramuscularly on the band.

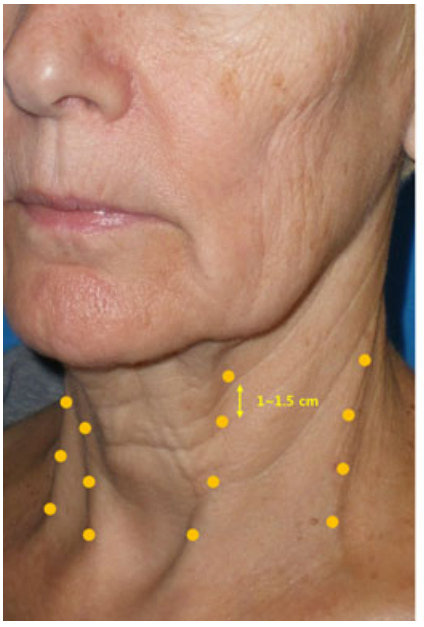
Ethnic Considerations The doses of BTA and injection points recommended by KACD are similar to those in Western recommendations.

Safety Concerns Dysphagia, dysphonia, and neck weakness are serious potential adverse events.

Calf Hypertrophy

Calf hypertrophy is a much-favored indication for BTA treatment in Asian women.

Injection Target, Points, Dose, and Technique The Achilles tendon connects the calf muscles, which have three heads arising from the two major muscle masses, to the foot. Three injections per portion



Platysmal bands	
Target muscle	• Platysma
Injection level	Intramuscular
Injection points	3~4 points /band
Injection dose	2 U
Total dose	24 ~ 32 U

Side effects	Prevention
Difficulty in swallowing (very rare)	Injection should be limited to the platysma muscle. Avoid injecting high doses.

Figure 17. Botulinum toxin injection to platysmal bands.



Calf contouring	
Target muscle	• Gastrocnemius
Injection level	Intramuscular
Injection points	12 points (6/side)
Injection dose	8 ~ 10 U
Total dose	100 ~ 120 U (50 ~ 60 U/side)

Side effects	Prevention
Difficulty in walking	Do not over-weaken the lateral gastrocnemius. Avoid injecting too high doses, too deeply (should spare the soleus which is involved in walking)

Figure 18. Botulinum toxin injection for calf contouring.

(6 points for each side of the leg) are recommended to reduce the sizes of the medial and lateral portions of the calf muscles (Figure 18). A total dose of 100 to 120 U (50–60 U per side) is recommended, divided between six-injection points (approximately 8–10 U per point). Injections should be intramuscular and perpendicular.

Ethnic Considerations In general, Asian women have shorter legs and thicker calves than Caucasians.

BTA treatment has been popular in Asia, with its ability to make the legs look long and slim. The mechanism of BTA treatment for calf hypertrophy is identical to that of benign masseter hypertrophy.

Safety Concerns Calf muscle impairments such as gait disturbance and fatigue after walking or running are the most common concerns after treatment, but they rarely occur.

Summary

The KACD consensus recommendations have addressed questions regarding treatment with BTA in general and specific common indications on the face, neck, and calf in Asians. For each indication, the anatomy of the target muscles was briefly reviewed, and the recommended injection points, dose, and injection techniques were provided (Table 1). This paper also reviewed the complications that may follow BTA injection in Asians to ensure maximal treatment safety. Most importantly, the KACD paper has made extensive reviews on ethnic differences, such as the anatomy of facial muscles and ideas of beauty. Indications for which there are major discrepancies between the KACD consensus recommendations and the latest western recommendations are summarized in Table 2. The KACD consensus recommendations will help ensure treatment safety and efficacy with BTA in Asians and can be further adapted in clinical practice to meet individual needs.

References

1. Scott AB. Botulinum toxin injection into extrocular muscles as an alternative to strabismus surgery. J Pediatr Ophthalmol Strabismus 1980;17:21–5.
2. Carruthers JD, Carruthers JA. Treatment of glabellar frown lines with C. botulinum-A exotoxin. J Dermatol Surg Oncol 1992;18:17–21.
3. Carruthers J, Fagien S, Matarasso SL; Botox Consensus Group. Consensus recommendations on the use of botulinum toxin type A in facial aesthetics. Plast Reconstr Surg 2004; 114:1–22S.
4. Rzany B. Bericht zum. Expertentreffen zur Anwendung von Botulinumtoxin A in der Ästhetischen Dermatologie. Kosmetische Medizin 2003;24:2–8.
5. Rzany B, Fratila A, Heckmann M. Expertentreffen zur Anwendung von Botulinumtoxin A in der Ästhetischen Dermatologie. Kosmetische Medizin 2005;3:1–8.
6. Carruthers JD, Glogau RG, Blitzer A; Facial Aesthetics Consensus Group Faculty. Advances in facial rejuvenation: botulinum toxin type A, hyaluronic acid dermal fillers, and combination therapies—consensus recommendations. Plast Reconstr Surg 2008; 121:5–30S.
7. Kane M, Donofrio L, Ascher B, Hexsel D, et al. Expanding the use of neurotoxins in facial aesthetics: a consensus panel's assessment and recommendations. J Drugs Dermatol 2010;9:s7–22.
8. Ascher B, Talarico S, Cassuto D, Escobar S, et al. International consensus recommendations on the aesthetic usage of botulinum toxin type A (Speywood Unit)—Part I: upper facial wrinkles. J Eur Acad Dermatol Venereol 2010;24:1278–84.
9. Ascher B, Talarico S, Cassuto D, Escobar S, et al. International consensus recommendations on the aesthetic usage of botulinum toxin type A (Speywood Unit)—Part II: wrinkles on the middle and lower face, neck and chest. J Eur Acad Dermatol Venereol 2010;24:1285–95.
10. Rho NK, Kim HS, Kim YS, Kim YJ, et al. Botulinum toxin type A for facial wrinkles and benign masseter hypertrophy in Korean patients. Korean J Dermatol 2010;48:823–31.
11. Carruthers A, Carruthers J. Botulinum toxin products overview. Skin Therapy Lett 2008;13:1–4.
12. Karsai S, Raulin C. Current evidence on the unit equivalence of different botulinum neurotoxin A formulations and recommendations for clinical practice in dermatology. Dermatol Surg 2009;35:1–8.
13. Stone AV, Ma J, Whitlock PW, Koman LA, et al. Effects of Botox and Neuronox on muscle force generation in mice. J Orthop Res 2007;25:1658–64.
14. Tzou CHJ, Giovanoli P, Ploner M, Frey M. Are there ethnic differences of facial movements between Europeans and Asians? Br J Plast Surg 2005;58:183–95.
15. Tsukahara K, Fujimura T, Yoshida Y, Kitahara T, et al. Comparison of age-related changes in wrinkling and sagging of the skin in Caucasian females and in Japanese females. J Cosmet Sci 2004;55:351–71.
16. Choe K, Sclafani A, Litner J, Yu GP, et al. The Korean American woman's face: anthropometric measurements and quantitative analysis of facial aesthetics. Arch Facial Plast Surg 2004;6: 244–52.
17. Ozsoy Z, Genc B, Gozu A. A new technique applying botulinum toxin in narrow and wide foreheads. Aesthetic Plast Surg 2005;29:368–72.
18. Wen JH, Ji ZL, Lu F, et al. Clinical anatomical study of the corrugator and procerus muscles. China J Aesthetic Plast Surg 2001;12:42–5.
19. de Almeida ART, da Costa Marques ERM, Kadunc BV. Glabellar wrinkles: a pilot study of contraction patterns. Surg Cosmet Dermatol 2010;2:23–8.
20. Rho NK. Ethnic considerations for the facial aesthetic uses of botulinum toxin- an Asian perspective. Kosmetische Medizin 2011;6:216–24.
21. Biller JA, Kim DW. A contemporary assessment of facial aesthetic preferences. Arch Facial Plast Surg 2009;1:91–7.

22. Tsukahara K, Sugata K, Osanai O, Ohuchi A, et al. Comparison of age-related changes in facial wrinkles and sagging in the skin of Japanese, Chinese and Thai women. J Dermatol Sci 2007;47:19–28.

23. Lee Y, Hwang K. Skin thickness of Korean adults. Surg Radiol Anat 2002;24:183–9.

24. Carruthers J, Carruthers A. Aesthetic botulinum A toxin in the mid and lower face and neck. Dermatol Surg 2003;29:468–76.

25. Churchill Living Stone. Gray's anatomy (Chapter 32, 40th ed). Edinburgh: Churchill Livingstone, 2008, pp. 547–59.

26. Kim NH, Chung JH, Park RH, Park JB. The use of botulinum toxin type A in aesthetic mandibular contouring. Plast Reconstr Surg 2005;115:919–30.

27. Kim KS, Byun YS, Kim YJ, Kim ST. Muscle weakness after repeated injection of botulinum toxin type A evaluated according to bite force measurement of human masseter muscle. Dermatol Surg 2009;35:1902–6.

Address correspondence and reprint requests to: Hei Sung Kim, MD, Department of Dermatology, Incheon St. Mary's Hospital, Catholic University of Korea, Dongsuro, Bupyeong-gu, 150-713, Incheon, Korea, or e-mails: hazelkimhoho@gmail.com; hazelkimhoho@catholic.ac.kr